Numerical Analysis of Lid-Driven Cavity Flow

Vorticity–streamfunction formulation and primary-vortex characteristics

1. Aim

The objective of this experiment is to obtain the incompressible flow field inside a square cavity driven by the motion of its upper wall. The primary recirculation is examined through the vorticity and streamfunction fields, together with velocity distributions along the cavity centre lines.

2. Computational Model and Parameters

A two-dimensional Cartesian domain is considered. The upper boundary moves at a prescribed horizontal speed, while the bottom and side boundaries remain stationary. The flow is computed using the vorticity–streamfunction formulation for Reynolds numbers 100 and 400.

Parameter	Specification
Coordinate system	Cartesian
Grid size	65 × 65
Reynolds numbers	100 and 400
Lid velocity, U_lid	2 (listed in parameter table)
CFL number	0.8
Tolerance	1e-0.8 (as written in source)
Maximum time steps	2000
SOR relaxation factor	1.7

3. Governing Formulation and Boundary Conditions

The vorticity–streamfunction approach is used to represent the two-dimensional incompressible flow. The streamfunction is obtained from a Poisson equation relating it to vorticity, while the velocity components are calculated from derivatives of the streamfunction. Vorticity is then advanced using its transport equation.

The computational region is a unit square. The top wall moves horizontally at the specified lid speed, whereas the bottom and vertical walls are stationary. The streamfunction is set to zero along all cavity walls. Wall vorticity is imposed using a Thom-type boundary condition.

4. Numerical Method

Vorticity transport is discretized with an explicit upwind scheme. At each step, the streamfunction Poisson equation is solved using red-black Successive Over-Relaxation (SOR). The velocity field is recovered from the streamfunction, and the wall-vorticity values are updated using the Thom-type relation.

5. Quantities Used for Assessment

The main diagnostics are the primary-vortex centre, minimum streamfunction, horizontal and vertical centreline velocity profiles, change in vorticity between iterations, and the Poisson-solver residual. The computed flow is also compared with the benchmark values quoted in the source report.

6. Results for Reynolds Numbers 100 and 400

The following table reorganizes the numerical values reported in the source. The benchmark entries are retained as quoted so that the comparison remains consistent with the supplied report.

Quantity	Re = 100	Re = 400
Grid / maximum steps	65 × 65 / 2000	65 × 65 / 2000
Primary-vortex centre (x, y)	(0.63, 0.77)	(0.66, 0.70)
Minimum streamfunction	–0.096	–0.090
Minimum u at x = 0.5	–0.185 at y ≈ 0.55	–0.205 at y ≈ 0.48
u = 0 crossing on vertical centreline	y ≈ 0.78	y ≈ 0.70
Maximum v at y = 0.5	+0.123 at x ≈ 0.23	+0.129 at x ≈ 0.39
Minimum v at y = 0.5	–0.182 at x ≈ 0.82	–0.252 at x ≈ 0.83
v = 0 crossing on horizontal centreline	x ≈ 0.57	x ≈ 0.60
Vorticity-change norm at step 2000	2.3 × 10^(-4)	4.7 × 10^(-4)
Benchmark minimum streamfunction	–0.103 at (0.62, 0.73)	–0.114 at (0.55, 0.61)
Benchmark minimum u on vertical centreline	–0.21 at y ≈ 0.45	–0.33 at y ≈ 0.28

7. Flow-Field and Centreline Behaviour

The streamfunction contours show a dominant clockwise recirculation beneath the moving lid. The vortex centre is reported at approximately (0.63, 0.77) for Re = 100 and (0.66, 0.70) for Re = 400. The minimum streamfunction values are –0.096 and –0.090, respectively, indicating a weaker primary vortex at the higher Reynolds number in the computed results.

The horizontal centreline profiles show upward flow on the left side and downward flow on the right side of the cavity. At Re = 400, the downward velocity minimum becomes more negative, reaching approximately –0.252 compared with –0.182 at Re = 100. The vertical centreline profiles also show reverse flow, with the minimum u-velocity changing from approximately –0.185 to –0.205.

8. Convergence and Benchmark Comparison

The vorticity-change norm decreases rapidly during the initial iterations and then decays more gradually. By step 2000 it reaches approximately 2.3 × 10^(-4) for Re = 100 and 4.7 × 10^(-4) for Re = 400. The curves are still decreasing at the end of the run, and the source states that neither case reached the specified tolerance. Both calculations therefore represent near-steady states rather than fully converged steady solutions. The Poisson residual is not plotted, so the accuracy of the SOR solution cannot be assessed from the supplied results.

Compared with the benchmark values quoted from Ghia et al., the Re = 100 solution is in qualitative agreement, although the computed vortex is weaker. The discrepancy is larger at Re = 400: the computed minimum streamfunction is less negative than the benchmark, the vertical-centreline velocity minimum is under-predicted, and the vortex centre is displaced from the benchmark location.

9. Sources of Numerical Discrepancy

The source report identifies first-order upwind discretization as a likely source of numerical diffusion, which can weaken the primary vortex. The 65 × 65 mesh may also be too coarse to resolve the stronger gradients at higher Reynolds number. In addition, the runs terminate at the step limit before reaching the stated convergence tolerance. The first-order wall-vorticity treatment may further limit accuracy near the moving lid. Secondary corner vortices are not clearly resolved in the supplied streamfunction contours.

10. Conclusion

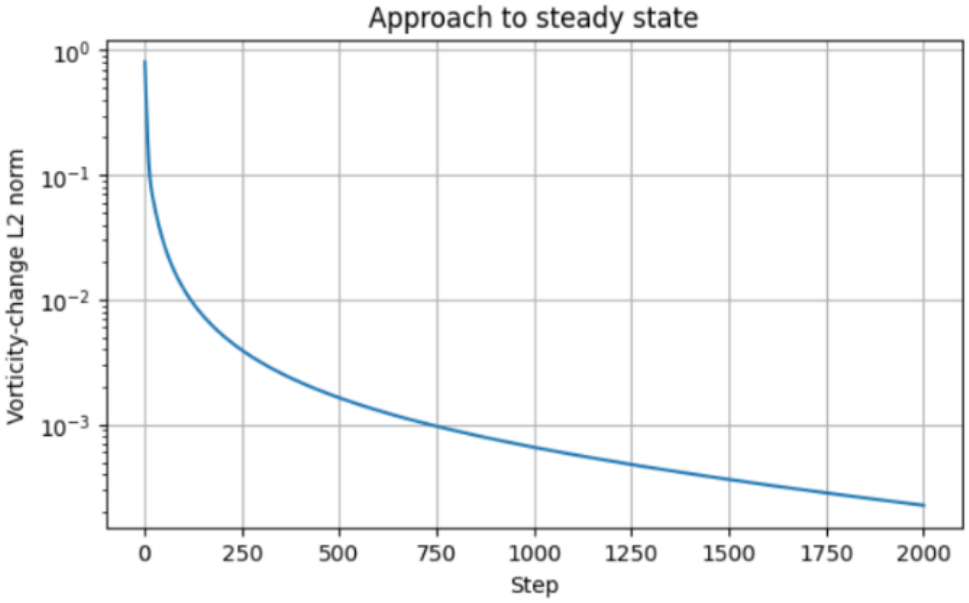
The lid-driven cavity problem was simulated using a vorticity–streamfunction formulation with explicit upwind vorticity transport, red-black SOR for the Poisson equation and Thom-type wall-vorticity conditions. The computed flow exhibits the expected primary clockwise vortex and corresponding centreline reverse-flow patterns at Reynolds numbers 100 and 400.

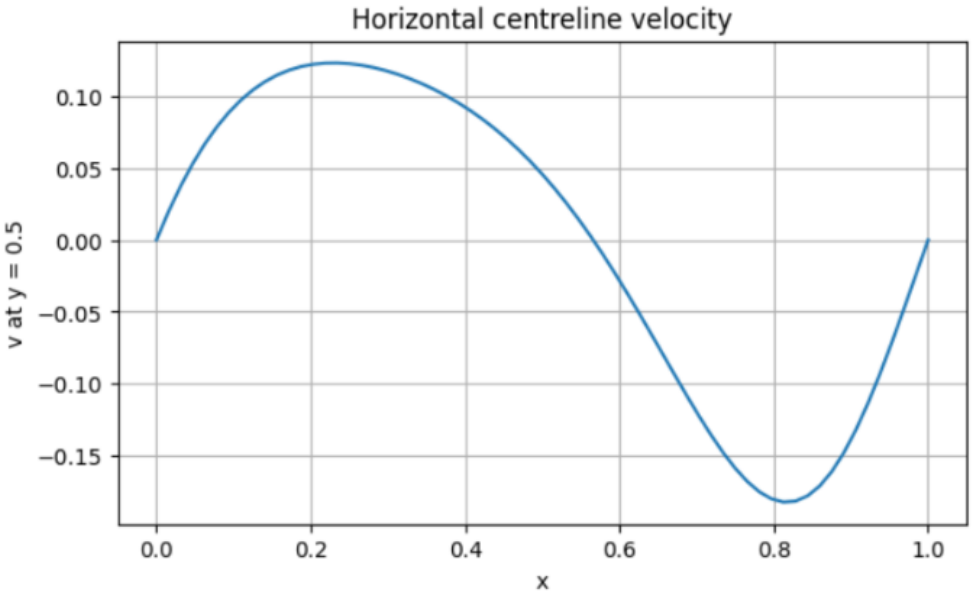
The results at Re = 100 compare more closely with the quoted benchmark than those at Re = 400. Quantitative differences at the higher Reynolds number, together with the continuing decrease in the vorticity-change norm, indicate that the present calculations are affected by numerical diffusion, grid resolution and incomplete convergence. A finer mesh, a higher-order convective scheme, longer runs to the convergence criterion and monitoring of the Poisson residual would improve the reliability of the solution.

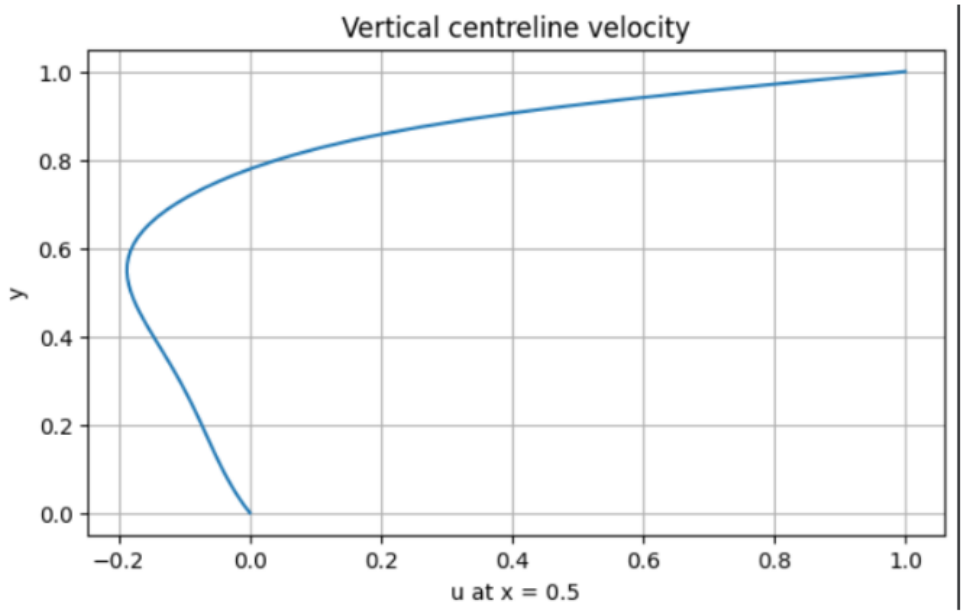
11. Original Simulation Figures

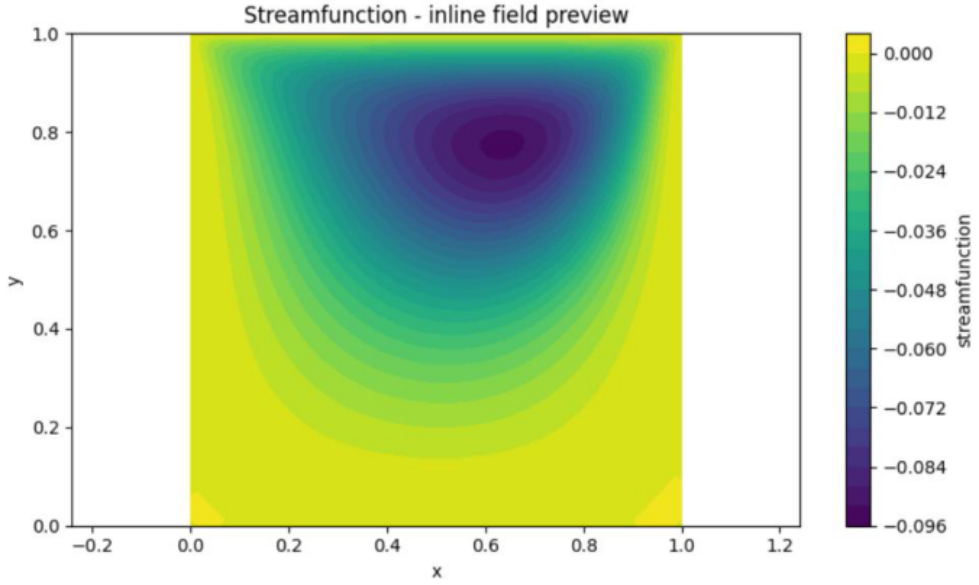
The original convergence histories, centreline velocity profiles and streamfunction contours are reproduced in the following pages.

For Raynolds number 100:

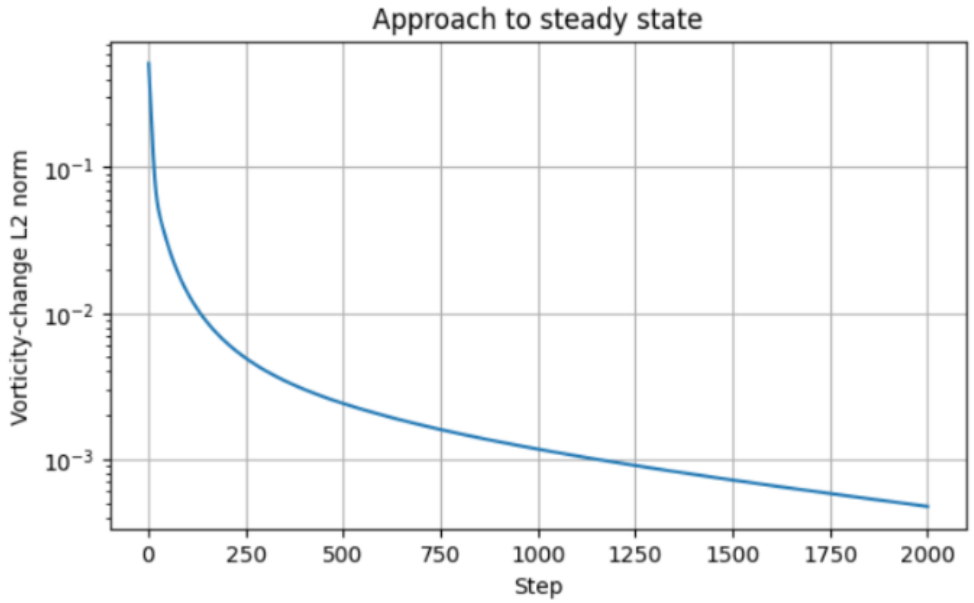


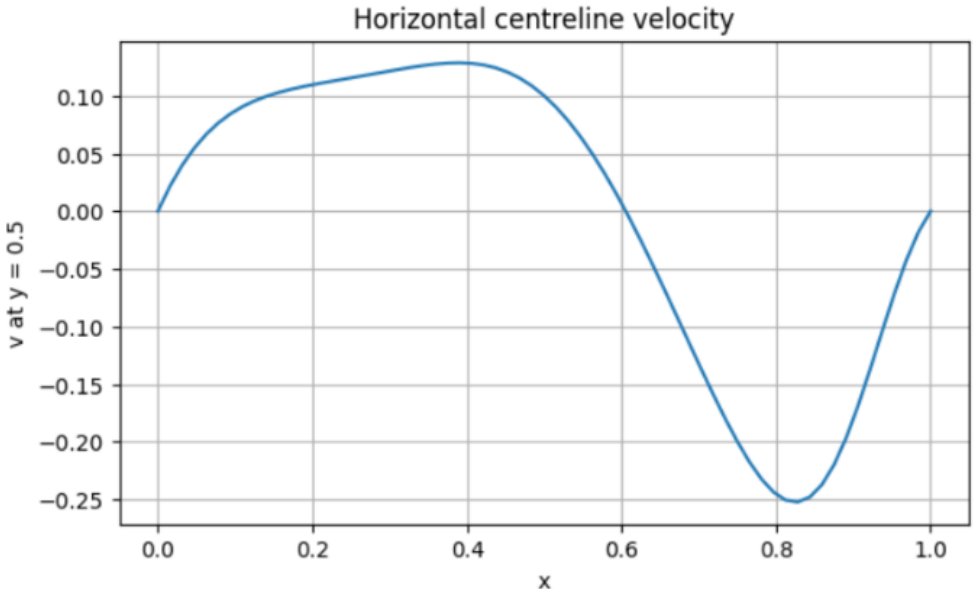


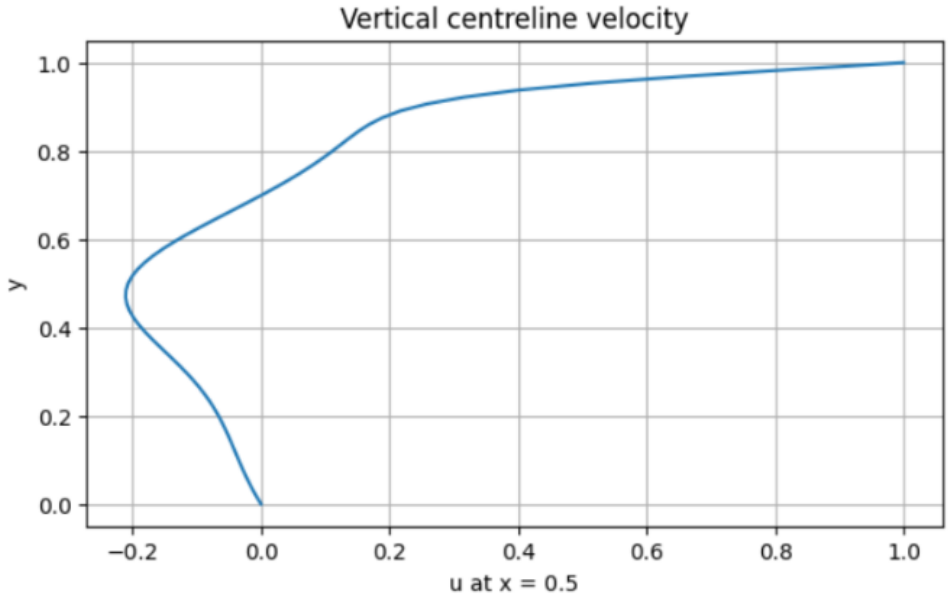

Vertical centreline velocity
1.0
0.8
0.6
0.4
0.2
0.0
y
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
u at x = 0.5

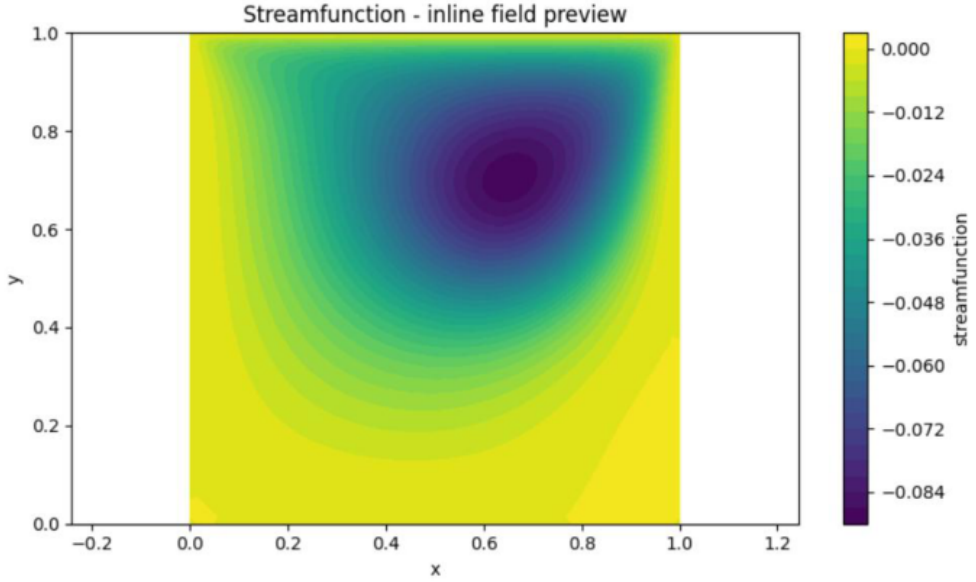

Streamfunction - inline field preview
1.0
0.8
0.6
0.4
0.2
0.0
y
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
1.2
x
0.000
−0.012
−0.024
−0.036
−0.048
−0.060
−0.072
−0.084
−0.096
streamfunction

For Raynolds number 400:






Vertical centreline velocity
1.0
0.8
0.6
0.4
0.2
0.0
y
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
u at x = 0.5


Streamfunction - inline field preview
1.0
0.8
0.6
0.4
0.2
0.0
y
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
1.2
x
0.000
−0.012
−0.024
−0.036
−0.048
−0.060
−0.072
−0.084
streamfunction

Results:

Parameter	**Re = 100**	**Re = 400**
Grid / steps run	65 × 65 / 2000 (max)	65 × 65 / 2000 (max)
Primary vortex centre (x, y)	≈ (0.63, 0.77)	≈ (0.66, 0.70)
Minimum streamfunction, ψ_min	≈ −0.096	≈ −0.090
Min. u on vertical centreline (x = 0.5)	≈ −0.185 at y ≈ 0.55	≈ −0.205 at y ≈ 0.48
u = 0 crossing on vertical centreline	y ≈ 0.78	y ≈ 0.70
Max. v on horizontal centreline (y = 0.5)	≈ +0.123 at x ≈ 0.23	≈ +0.129 at x ≈ 0.39
Min. v on horizontal centreline (y = 0.5)	≈ −0.182 at x ≈ 0.82	≈ −0.252 at x ≈ 0.83
v = 0 crossing on horizontal centreline	x ≈ 0.57	x ≈ 0.60
Vorticity-change norm, step 0	$\approx 8 \times 10^{-1}$	$\approx 5 \times 10^{-1}$
Vorticity-change norm, step 250	$\approx 4 \times 10^{-3}$	$\approx 5 \times 10^{-3}$
Vorticity-change norm, step 2000	$\approx 2.3 \times 10^{-4}$	$\approx 4.7 \times 10^{-4}$
Benchmark ψ_min, vortex centre (Ghia et al., ≈)	≈ −0.103 at (0.62, 0.73)	≈ −0.114 at (0.55, 0.61)
Benchmark min. u on vertical centreline (≈)	≈ −0.21 at y ≈ 0.45	≈ −0.33 at y ≈ 0.28

Conclusion:

The lid-driven cavity flow in a unit square was solved with the vorticity–streamfunction formulation, using explicit upwind vorticity transport, red-black SOR for the streamfunction Poisson equation and Thom-type wall vorticity, on a 65 × 65 grid at Reynolds numbers of 100 and 400. In both cases a single primary vortex forms below the moving lid, with the streamfunction negative (clockwise circulation) in the vortex core. The primary vortex centre lies at approximately

(0.63, 0.77) for Re = 100 and (0.66, 0.70) for Re = 400, with minimum streamfunction of about −0.096 and −0.090 respectively.

Centreline velocities: On the vertical centreline the u-velocity rises from zero at the bottom wall, reaches a reverse-flow minimum of about −0.185 (Re = 100, y ≈ 0.55) and −0.205 (Re = 400, y ≈ 0.48), crosses zero at y ≈ 0.78 and 0.70, and reaches the lid speed of 1 at y = 1. On the horizontal centreline the v-velocity is positive on the left (up to ≈ 0.123 at x ≈ 0.23 for Re = 100 and ≈ 0.129 at x ≈ 0.39 for Re = 400) and negative on the right (down to ≈ −0.182 and ≈ −0.252 at x ≈ 0.82–0.83), which is the expected upward flow near the left wall and downward flow near the right wall for a clockwise vortex. The profiles are less symmetric and the right-hand downward jet is stronger at Re = 400, and the shear layer under the lid is thinner: u = 0.6 is reached at y ≈ 0.945 for Re = 100 and at y ≈ 0.965 for Re = 400. This is consistent with convection becoming more important relative to viscous diffusion as Re increases.

Convergence: The vorticity-change norm falls quickly from about 0.8 (Re = 100) and 0.5 (Re = 400) to about 4–5 × 10^{-3} by step 250, but then decays only slowly, reaching about 2.3 × 10^{-4} and 4.7 × 10^{-4} at step 2000. The curves are still falling at the end of the run and no plateau is visible, so neither case reached the specified tolerance and both runs stopped at the step limit. The solution is therefore a close-to-steady transient state and not a fully converged steady state. The vorticity-change norm at step 2000 is about twice as high at Re = 400 as at Re = 100, so the higher-Re case is further from steady state. The Poisson residual was not plotted, so the accuracy of the SOR solve cannot be assessed from the results.

Comparison with benchmark data: The well-known benchmark solution of Ghia et al. for the same problem gives a minimum streamfunction of about −0.103 at (0.62, 0.73) for Re = 100 and about −0.114 at (0.55, 0.61) for Re = 400, and the minimum u on the vertical centreline is about −0.21 at Re = 100 and about −0.33 at Re = 400. The present results are in qualitative agreement at Re = 100, with the vortex centre and the u-minimum close to the benchmark, but the vortex strength is lower by about 7%, and the velocity extrema are lower. At Re = 400 the differences are larger: the streamfunction minimum is about 20% too weak, the u-minimum of about −0.21 is far from about −0.33, and the vortex centre at (0.66, 0.70) lies to the right of and above the benchmark position. The benchmark vortex moves towards the cavity centre as Re increases, whereas here it moves right. The velocity extrema are therefore under-predicted at both Reynolds numbers, and more so at Re = 400.

Likely causes of the discrepancy: First-order upwind transport of vorticity introduces numerical diffusion that acts like an increased viscosity, which weakens the vortex and thins the effective Reynolds number, and this effect grows with Re on a coarse 65 × 65 grid. The first-order Thom wall-vorticity boundary condition also limits accuracy near the lid. In addition, the runs were stopped before steady state, so the vortex is still adjusting. Grid dependence was listed in the observations but only the 65 × 65 grid is shown, so the grid effect could not be separated from the scheme effect. Secondary corner vortices were not resolved; the faint patches near the bottom

corners of the streamfunction plots are too weak to be identified as vortices, and at Re = 400 they are expected to be very small and need a finer grid and a converged solution to appear.

Overall, the method reproduces the qualitative structure of the lid-driven cavity flow: a single clockwise primary vortex below the lid, reverse flow along the lower part of the vertical centreline, a thinner lid shear layer and stronger asymmetry at higher Re. However, the quantitative accuracy is limited, especially at Re = 400, by first-order upwind diffusion, the coarse grid and incomplete convergence. Running to the specified tolerance, plotting the Poisson residual, refining the grid (e.g. 129 × 129 and above), and using a second-order convective scheme would be needed for results suitable for quantitative comparison with the benchmark. The parameter table also lists a lid speed of 2 whereas the plots show a lid speed of 1.